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The company receives $100 * \$10.00$, or $\$1,000$, in proceeds. Its share price is $\$20.00$, so it can repurchase 50 shares with these proceeds. There are now 50 net additional shares outstanding.

You add the 50 RSUs as if they were common shares, so now there's a total of 100 additional shares outstanding.

The company's share price of $\$20.00$ exceeds the conversion price of $\$10.00$, so the convertible bonds can convert into shares.

Divide the par value by the conversion price to determine the shares per bond:

$$\$100 / \$10.00 = 10 \text{ new shares per bond}$$

There are 100 individual convertible bonds, so they create $100 * 10 = 1,000$ new shares.

These changes create 1,100 additional shares, so the diluted share count is now 11,100, and the Diluted Equity Value is $11,100 * \$20.00$, or $\$222,000$.

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Valuation Methodologies

These questions are high-level, but many candidates don't understand *the point* of valuation.

You can't answer the advanced questions without knowing that, so don't dismiss this category as "too basic" – even if you have significant work experience.

There are dozens of valuation methodologies, but in real-life interviews, bankers tend to focus on **comparable public companies**, **precedent transactions**, and the **DCF**.

This section and the [next one](#) cover the first two categories, and [the two sections after that cover the DCF](#).

- [Interview Guide – Valuation](#) | [Quiz Questions](#)
- [Core Financial Modeling – Valuation Module](#)

• Company Value =

Cash Flow

$\frac{\text{Cash Flow}}{(\text{Discount Rate} - \text{Cash Flow Growth Rate})}$

Where the Cash Flow Growth Rate
Must Be $<$ Discount Rate

- **Valuation is more than this formula because the Discount Rate and Cash Flow Growth Rate *change* over time**



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1. Public companies already have Market Caps and Share Prices. Why do you need to “value them” at all? You already know how much they’re worth.

Because a company’s Market Cap and Share Price reflect its **Current Value** according to “the market as a whole” – but the market might be wrong!

You value companies to see if the market’s views are correct and whether a company’s value *might change* based on your views.

It’s like going home shopping, finding a house with a list price of \$500K, but then negotiating a lower price because *you believe* it is worth only \$450K based on your research.

2. What are the advantages and disadvantages of the 3 main valuation methodologies?

Public Comps are useful because they’re based on real market data, are quick to calculate and explain, and do not depend on far-in-the-future assumptions.

However, there may not be truly comparable companies, the analysis will be less accurate for volatile or thinly traded companies, and it may undervalue companies’ long-term potential.

Precedent Transactions are useful because they’re based on the **real prices** that companies have paid for other companies, and they may better reflect industry trends than Public Comps.

However, the data is often spotty and misleading, there may not be truly comparable transactions, and specific deal terms and market conditions might distort the multiples.

DCF Analysis is the most “correct” methodology according to finance theory, it’s less subject to market fluctuations, and it better reflects company-specific factors and long-term trends.

However, it’s also very dependent on far-in-the-future assumptions, and there’s disagreement over the proper calculations for key figures like the Cost of Equity and WACC.

3. Which of the 3 main methodologies will produce the highest Implied Values?

This is a trick question because almost any methodology *could* produce the highest Implied Values depending on the industry, time period, and assumptions.

That said, Precedent Transactions often produce higher Implied Values than Public Comps because of the **control premium** – the extra amount that buyers must pay to acquire sellers.



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But it's tough to say how a DCF compares because it's far more dependent on the long-term assumptions used.

The safest answer is: "A DCF tends to produce the most variable output since it's so dependent on your assumptions, and Precedent Transactions tend to produce higher values than the Public Comps because of the control premium."

4. Which one should be worth more: A \$500 million EBITDA healthcare company or a \$500 million EBITDA industrials company?

Assume the growth rates and margins are the same.

In all likelihood, the healthcare company will be worth more because healthcare is a less asset-intensive industry. That means the company's CapEx and Working Capital requirements will be lower, and its cash flow will be higher.

Healthcare, at least in some sectors, also tends to be more of a "growth industry" than industrials.

The Discount Rate might also be higher for the healthcare company, but the lower asset intensity and higher expected growth rates could offset that.

This answer is an *extreme* generalization, so you would need more information to give a detailed answer.

5. Can you walk me through how you use Public Comps and Precedent Transactions in a valuation?

First, you select the companies and transactions based on industry, size, and geography (and time for the transactions).

Then, you determine the appropriate metrics and multiples for each set – for example, revenue, revenue growth, EBITDA, EBITDA margins, and revenue and EBITDA multiples – and calculate them for all the companies and transactions.

Next, you calculate the minimum, 25th percentile, median, 75th percentile, and maximum for each valuation multiple in the set.

Finally, you apply these numbers to the financial metrics of the company you're analyzing to estimate its Implied Value.



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For example, if the company you're valuing has \$100 million in LTM EBITDA, and the median LTM TEV / EBITDA multiple in a set of comparable companies is 7x, then the company's implied Enterprise Value is \$700 million.

You then calculate its Implied Value for all the other multiples to get a range of possible values.

6. Can you give a few examples of how you might screen for "similar" Comparable Public Companies and Precedent Transactions?

You screen based on **geography**, **industry**, and **size** (and **time** for Precedent Transactions).

Here are a few example screens:

- **Comparable Company Screen:** U.S.-based steel manufacturing companies with over \$500 million in revenue.
- **Comparable Company Screen:** European legacy airlines with over €1 billion in EBITDA.
- **Precedent Transaction Screen:** Latin American M&A transactions over the past 3 years involving consumer/retail sellers with over \$1 billion USD in revenue.
- **Precedent Transaction Screen:** Australian M&A transactions over the past 2 years involving infrastructure sellers with over \$200 million AUD in revenue.

7. How do you decide which metrics and multiples to use in these methodologies?

You usually look at a sales-based metric and its corresponding multiple and 1-2 profitability-based metrics and their multiples. For example, you might use Revenue, EBITDA, and Net Income and their corresponding multiples: TEV / Revenue, TEV / EBITDA, and P / E.

You do this because you want to value a company in relation to how much it sells and how much it *keeps* from those sales.

Sometimes, you'll drop the sales-based multiples and focus on the profitability or cash flow-based ones (EBIT, EBITDA, Net Income, Free Cash Flow, etc.).

8. Why do you look at BOTH historical and projected metrics in these methodologies?



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Historical metrics are useful because they're based on what happened in real life, but they can also be deceptive if there were non-recurring items or if the company made acquisitions or divestitures.

Projected metrics are useful because they assume the company will operate in a "steady state" without acquisitions, divestitures, or non-recurring items, but they're also less reliable because they're based on predictions rather than historical events.

9. When calculating the forward multiples for the comparable companies, should you use each company's Current Equity Value or Current Enterprise Value, or should you project them to get the Year 1 or Year 2 values?

You always use the Current Equity Value or Current Enterprise Value. **NEVER "project" either one.**

A company's share price, and, therefore, both Current Equity Value and Current Enterprise Value, reflects past performance and future expectations.

So, to "project" these metrics, you'd have to jump into the future and see what future expectations are at *that* point in time and then time travel back to the present.

10. How do you interpret the Public Comps? What does it mean if the median multiples are above or below the ones of the company you're valuing?

The interpretation depends on how your company's growth rates and margins compare to those of the comparable companies.

Public Comps are most meaningful when the growth rates and margins are **similar**, but the multiples are **different**. This could mean that the company you're valuing is mispriced.

For example, maybe all the companies are growing at 10 – 15%, and they all have EBITDA margins of 10 – 15%. Your company also has growth rates and margins in these ranges.

However, your company trades at TEV / EBITDA multiples of 6x to 8x, while the comparable companies all trade at multiples of 10x to 12x.

This result could indicate that your company is **undervalued** since its multiples are lower, but its growth rates, margins, industry, and size are comparable.



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11. What is a Liquidation Valuation, and when is it useful and not so useful?

In a Liquidation Valuation, you value a company by estimating the market values of all its Assets, adding them up, and subtracting its Liabilities (i.e., you assume full repayment of all Liabilities based on the proceeds from the sale of all its Assets).

It gives you the company's **Implied Equity Value** because you're valuing its *Net Assets*, not its Net Operating Assets.

This methodology is useful for distressed companies because it tells you how much they might be worth if they liquidate and how much different lender groups might receive.

It's less useful for healthy, growing companies because it undervalues them significantly; assets like Net PP&E are always worth more to "going concern" companies.

12. How does a Dividend Discount Model (DDM) differ from a DCF?

In a DDM, rather than projecting Free Cash Flow, you project the company's **Dividends**, usually based on a per-share figure or a percentage of Net Income. You then discount the Dividends to their Present Value using the Cost of Equity and add them up.

To calculate the Terminal Value, you use an Equity Value-based multiple such as P / E, and you discount it to Present Value using the Cost of Equity.

You add the PV of the Terminal Value to the PV of the Dividends to calculate the company's Implied Equity Value rather than its Implied Enterprise Value (there's no "bridge"), and you divide it by the diluted share count to get the company's Implied Share Price.

The DDM is essential in some industries, such as commercial banks and insurance, useful in other industries that pay regular dividends, such as utilities, and not so useful for most others.

13. Why might you use an M&A Premiums analysis to value a company?

The M&A Premiums analysis applies only to **public companies** because you look at acquisitions of similar public companies and calculate the "premium" each buyer paid for each seller.

For example, if the seller's share price was \$12.00 before the deal, and the buyer paid \$15.00 per share, that represents a 25% premium.

You then use these percentages to value your company. If the median premium in a set of deals is 20%, and your company's share price is \$10.00, it's worth \$12.00 per share.



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This analysis is typically a supplement to Precedent Transactions and gives you another way to value your company besides the standard multiples. But it's also limited because M&A Premiums cannot indicate that a company is currently *undervalued*.

14. What are the advantages and disadvantages of a Sum-of-the-Parts Valuation?

The Sum-of-the-Parts methodology, in which you value each division of a company separately and add them to determine the company's Implied Value, works well for conglomerates that have *very* different divisions (e.g., retail vs. transportation vs. digital media segments).

The divisions operate in such different industries that it would be meaningless to value the company as a whole – no other public company would be comparable.

But Sum-of-the-Parts also takes far more time and effort to set up because you must find comparable companies and transactions for **each division**, build a separate DCF for each division, and so on.

Also, you might not have enough information to use it. Companies sometimes don't disclose EBIT, CapEx, or Working Capital by division, and they may not disclose the corporate overhead expenses that must be counted in the final step.

15. How do you set up an LBO valuation, and when is it useful?

You set up the LBO valuation by creating a [leveraged buyout model](#) in which a private equity firm acquires a company using Debt and Equity, holds it for several years, and then sells it for a certain multiple of EBITDA.

Most private equity firms target an internal rate of return (IRR) in a specific range, so you work backward and determine the **maximum price** the PE firm could pay to achieve a targeted IRR.

You could use the "Goal Seek" function in Excel to do this, and you solve for the purchase price based on constraints for the IRR, exit multiple, and Debt / Equity split.

This methodology is most useful for screening LBO candidates; it can also help a company understand what PE firms vs. normal companies might pay for it.

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